CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program .1

Aim

To write a Python program to solve the 8-Puzzle problem using the A Search Algorithm*.

2. Algorithm

1.Start the program.

2.Initialize the starting state and goal state of the 8-Puzzle.

3.Calculate the heuristic value using Manhattan distance.

4.Apply the A* Search Algorithm to select the best state.

5.Find the empty space represented by 0.

6.Move the empty space up, down, left, or right whenever the move is valid.

7.Calculate the cost for each new state.

8.Repeat until the goal state is reached.

9.Print all intermediate states from the starting state to the goal state.

10.Stop the program.

untitled (3.14.8)

File Edit Format Run Options Window Help

```
        if state in visited:
            continue
        visited.add(state)

        zero = state.index(0)
        moves = [-3, 3, -1, 1]

        for move in moves:
            new_pos = zero + move

            if 0 <= new_pos < 9:
                if abs(zero // 3 - new_pos // 3) + \
                   abs(zero % 3 - new_pos % 3) != 1:
                    continue

                new_state = list(state)
                new_state[zero], new_state[new_pos] = \
                    new_state[new_pos], new_state[zero]
                new_state = tuple(new_state)

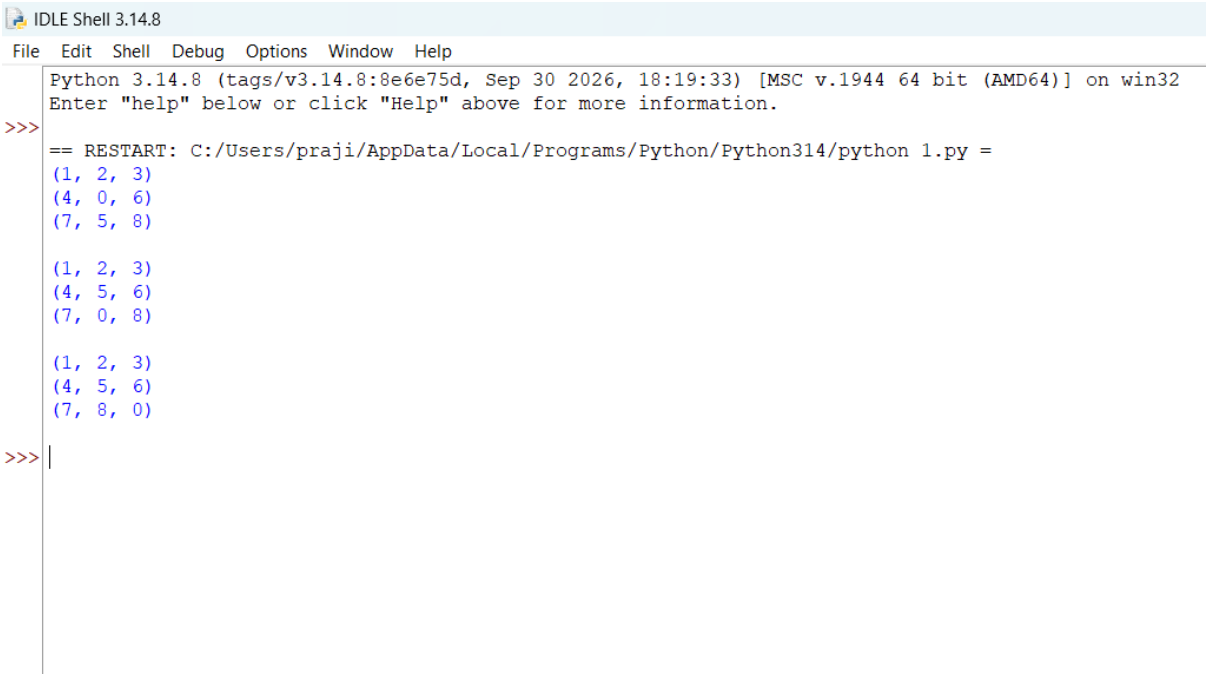
                if new_state not in visited:
                    parent[new_state] = state
                    heappush(pq, (
                        len_path(parent, new_state) +
                        heuristic(new_state),
                        new_state
                    ))

    return None

def len_path(parent, state):
    count = 0
    while parent[state] is not None:
        count += 1
        state = parent[state]
    return count

solution = solve(start)

if solution:
    for state in solution:
        for i in range(0, 9, 3):
            print(state[i:i+3])
        print()
else:
    print("No solution found")
```



3. Result

Thus, the Python program to solve the 8-Puzzle problem using the A Search Algorithm* was executed successfully, and the solution path from the initial state to the goal state was obtained.